

DeFi Super-App

Blockchain Technologies 2 — Final Project

Scenario A: AMM + ERC-4626 Vault + DAO Governance · Base Sepolia

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Smart Contracts & DeFi Primitives

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Governance & Oracles & Security

Dias Nygman

Frontend, Subgraph & DevOps

Project Overview

A full-stack DeFi protocol combining multiple primitives into one production-grade dApp.

Constant-Product AMM

$x \cdot y = k$, 0.3% fee, LP tokens, Yul sqrt

ERC-4626 Yield Vault

Tokenized shares, upgradeable UUPS

DAO Governance

Governor + Timelock, 4% quorum

Chainlink Oracle

Staleness checks, IOracle adapter

The Graph Subgraph

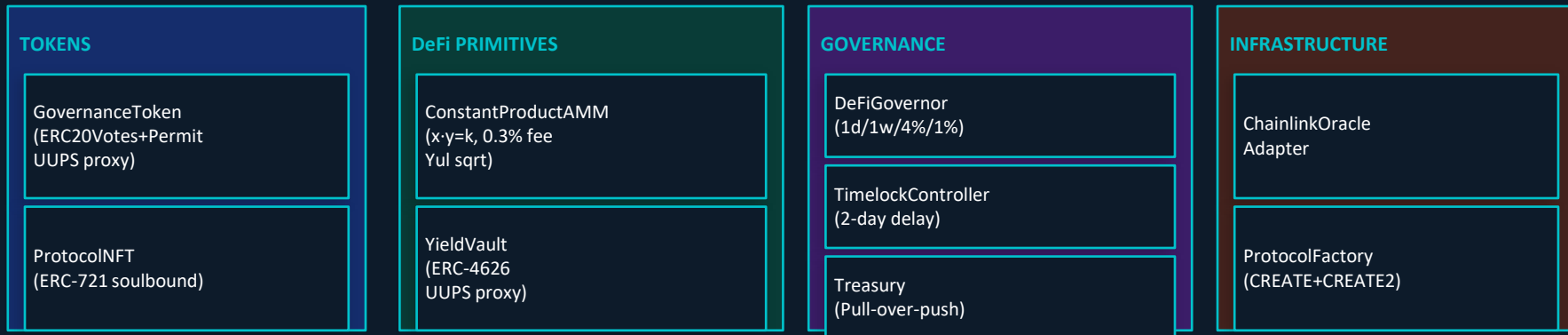
4 entities, GraphQL indexing

Base Sepolia L2

Verified contracts, 99.67% gas saving

System Architecture

C4 Level 2 — Contract Groups



External Dependencies



Governor is the sole PROPOSER on Timelock → no privileged actions outside governance flow

Factory uses CREATE (nonce-based) + CREATE2 (deterministic) for pool deployment

Smart Contracts Deep Dive

GovernanceToken

ERC20Votes + ERC20Permit

UUPS proxy · V1→V2 upgrade adds transfer tax

ConstantProductAMM

Built from scratch

Yul sqrt · CEI + nonReentrant · 0.3% fee

YieldVault

ERC-4626

UUPS proxy · all rounding invariants satisfied

DeFiGovernor

OZ Governor v5

1d delay / 1w period / 4% quorum / 1% threshold

TimelockController

OZ Timelock

2-day delay · controls treasury & upgrades

ProtocolNFT

ERC-721 soulbound

AccessControl MINTER_ROLE · non-transferable

ChainlinkOracleAdapter

IOracle interface

Staleness check · mock aggregator in tests

ProtocolFactory

Factory pattern

CREATE + CREATE2 · deterministic pool addresses

Treasury

Pull-over-push

claimETH / claimToken · no push-payment DoS

AMM & Vault Mechanics

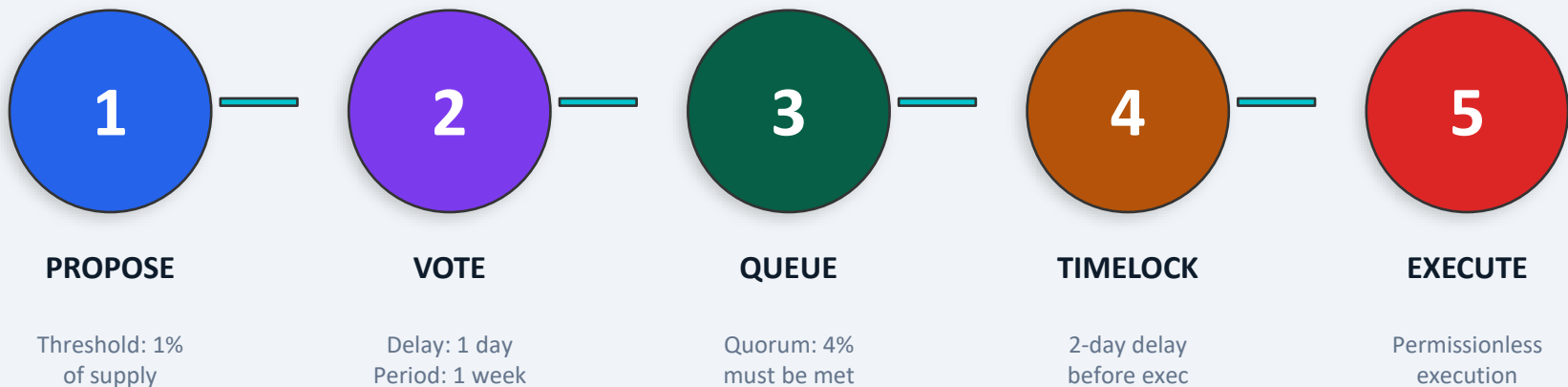
ConstantProductAMM

- ▶ Formula: $x \cdot y = k$ (constant product)
- ▶ 0.3% swap fee → LP earnings
- ▶ Slippage protection: amountOutMin check
- ▶ k-invariant verified after every swap
- ▶ LP tokens minted on addLiquidity
- ▶ Yul assembly sqrt (17% gas saving vs Solidity)
- ▶ CEI pattern + ReentrancyGuard on all writes
- ▶ SafeERC20 for all token transfers

YieldVault (ERC-4626)

- ▶ Deposit assets → receive vault shares
- ▶ Withdraw shares → receive assets + yield
- ▶ reportYield() credited to all holders
- ▶ Price-per-share = $\text{totalAssets} / \text{totalSupply}$
- ▶ ERC-4626 rounding invariants: all passing
- ▶ UUPS upgradeable — Timelock controls upgrade
- ▶ Chainlink oracle for yield rate sanity check
- ▶ AccessControl: YIELD_REPORTER_ROLE

DAO Governance Lifecycle



Governor Parameters

Voting Delay	1 day (7 200 blocks)
Voting Period	1 week (50 400 blocks)
Proposal Threshold	1% of supply
Quorum	4% of supply at snapshot
Timelock Delay	2 days

Security Guarantees

- ✓ Governor = sole PROPOSER on Timelock
- ✓ No flash-loan attack: votes use past block snapshot
- ✓ Whale protection: 4% quorum threshold
- ✓ Spam protection: 1% proposal threshold
- ✓ Upgrade/treasury only via Timelock

Security & Testing

80+

Total Tests

≥90%

Line Coverage

0

Slither High/Med

5

Design Patterns

Unit Tests — 50+

Every public/external function
including all revert paths

Fuzz Tests — 10+

AMM swap, vault deposit/withdraw,
governance voting power

Invariant Tests — 5+

k never decreases, supply conservation,
treasury accounting

Fork Tests — 3+

Real USDC, Uniswap V2 router,
Chainlink live feeds

Security Practices

✓ CEI + ReentrancyGuard on all writes

✓ SafeERC20 everywhere

✓ No tx.origin auth

(2000+ lines of final code / contract)

Gas Optimization & L1 vs L2

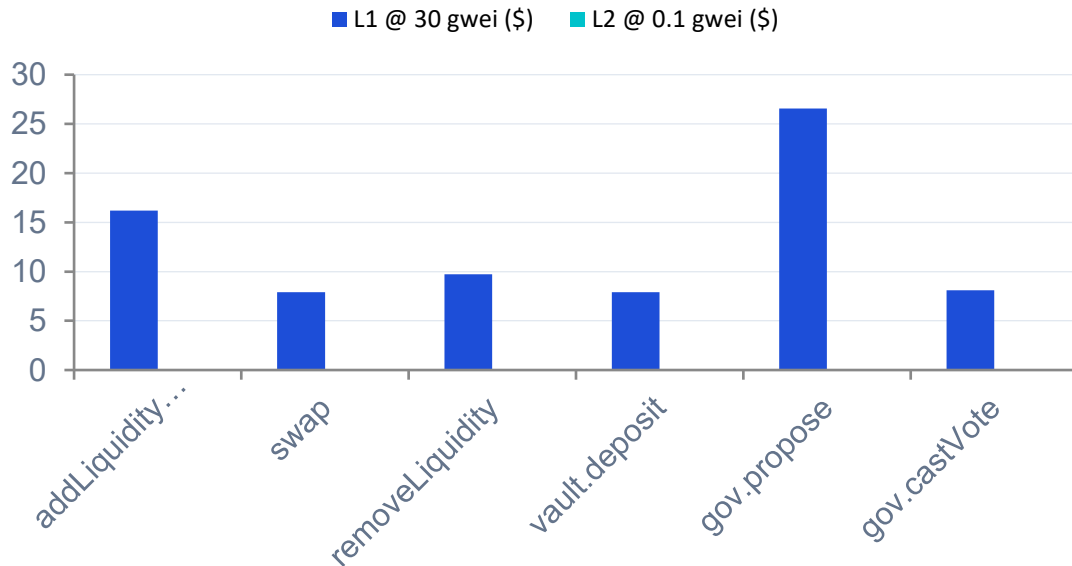
Yul vs Solidity sqrt

Yul (production) ~1 750 gas

Solidity equiv. ~2 050 gas

→ 17.1% saving

Transaction Cost: L1 vs L2 (\$USD)



Key Optimizations Applied

Optimization	Effect
Custom errors	No revert strings — saves calldata gas
Immutable addresses	token0/token1/feed — no SLOAD cost

Frontend & Subgraph

Frontend dApp

Tech stack:	React + Wagmi + Viem
Wallet:	MetaMask + WalletConnect
Reads:	Balance, voting power, delegates, pool reserves, vault shares
Writes:	Swap, deposit, castVote (+ 1 more)
Governance UI:	Proposal list with state badge + Vote button
Error handling:	Wrong network, rejected tx, insufficient balance — all user-friendly
Network detect:	Auto-prompts switch to Base Sepolia

The Graph Subgraph

Entities (4+ required):

- Pool
 - Swap
 - LiquidityEvent
 - VaultDeposit
 - VaultWithdraw
 - Proposal
 - Vote
-

Key GraphQL Queries (5+ documented):

- recentSwaps(first: N)
- poolStats(id)
- vaultHistory(account)
- activeProposals
- voterActivity(account)

Deployment & DevOps

Deployed Contracts — Base Sepolia (Chain ID 84532)

Contract	Address (Base Sepolia)
GovernanceToken	0xA2E174F9fAB0690489DC6EA4300BA242ee4A6807
ProtocolNFT	0xD98FC5f700E7D7059981Fc2Def353eE6ffe2a827
Oracle	0x38E6cBb4fb3F0a20C741A2Ed9A08d5e408C07faa
Factory	0xD07ccA3995FdE9a3933Fb8A761451Bc7f42Edd82
AMM Pool	0x995f482A876364d82313Bd3b0c29F1350d19188C
YieldVault	0x556234E9cBaC3B7a8d9254811a7aDaA098aE8533
Timelock	0xF08eaD1d04C585A050B817E694862E9B570214F2
Governor	0x99ca18C410840087ef301A38659fc090632d1De8
Treasury	0x0ED85c9c8E347C3Acc686A917E326708F0665C7

CI/CD Pipeline (GitHub Actions)

✓ forge build

✓ forge test (full suite)

✓ Slither analysis

✓ forge fmt --check + solhint

- ▶ deploy.s.sol — reproducible script
- ▶ Post-deploy verify.s.sol — checks owner = Timelock
- ▶ All contracts verified on BaseScan
- ▶ Conventional Commits enforced in CI

Team & Responsibilities

Alisher Sultanov

Smart Contracts & DeFi Primitives

- ConstantProductAMM — full implementation
- YieldVault (ERC-4626) + UUPS upgrade
- GovernanceToken V1 & V2 upgradeable
- ProtocolNFT (ERC-721 soulbound)
- ProtocolFactory (CREATE + CREATE2)
- Yul assembly sqrt + benchmark
- Unit & fuzz tests for DeFi primitives

Olzhas Elshibay

Governance · Oracles · Security

- DeFiGovernor + TimelockController
- Treasury (pull-over-push pattern)
- ChainlinkOracleAdapter + IOracle
- MockAggregator for testing
- Invariant & fork test suite
- Security audit report (8 pages)
- Vulnerability case studies (reentrancy, access-control)

Dias Nygman

Frontend · Subgraph · DevOps

- React dApp — Wagmi + Viem stack
- MetaMask + WalletConnect integration
- The Graph subgraph (7 entities)
- 5+ documented GraphQL queries
- GitHub Actions CI/CD pipeline
- deploy.s.sol + verify.s.sol scripts
- Architecture & gas optimization docs

What We Built

- ◆ 10 smart contracts · all verified on Base Sepolia
- ◆ Full DAO lifecycle: propose → vote → queue → execute
- ◆ Chainlink oracle with staleness guard + mock for tests
- ◆ React dApp · MetaMask + WalletConnect · error handling
- ◆ 8-page audit report + 6-page architecture document
- ◆ 80+ tests: unit, fuzz, invariant, fork · ≥90% coverage
- ◆ ERC-4626 vault with UUPS upgrade path V1→V2
- ◆ The Graph subgraph · 7 entities · 5 GraphQL queries
- ◆ GitHub Actions CI · Slither clean · Conventional Commits
- ◆ L2 gas savings: \$16 on L1 → \$0.05 on Base Sepolia

Thank you · Questions?